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Project Title:	HZE-induced skin carcinoma: an in-vivo-like multicellular epithelial model system for analysis of radiation quality effects and DSB repair processes		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:	For information purposes only--PI moved in June 2014; however, funding did not go with her. As of 6/1/14, Colorado State University, Department of Environmental and Radiological Health Sciences, 485 MRB - 1618 Campus Delivery, Fort Collins, CO 80523-1618, Office: (970) 491 7618, Email: Claudia.Wiese@colostate.edu		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
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Task Description:	There is considerable evidence that ionizing radiation causes non-melanoma skin cancer (NMSC), and the risk for NMSC in astronauts is significantly increased. To further assess the cancer risk for humans from space radiation, an in-vivo-like multicellular skin model system will be used. Multiple different endpoints as surrogate markers for skin cancer will be assessed. Persistent chromosomal damage, altered cell proliferation and evasion from apoptosis will be studied in 2D monolayer cultures of normal primary human keratinocytes exposed to a variety of charged nuclei (Aim 1). In addition, to more realistically recapitulate the complex architecture of a tissue, an in-vivo-like skin model system, composed of fibroblasts and keratinocytes, will be used to ascertain the risk of space radiation-induced cancer development processes in 3D (i.e. persistent chromosomal damage, altered cell proliferation, evasion from apoptosis and differentiation) (Aim 2). Last, a surface transplantation assay of exposed 3D reconstructed human epithelia onto immunodeficient mice will be carried out to test for intermediate and late events in cancer progression (cell invasion and metastasis, respectively) (Aim 3). We hypothesize that, in basal cells, unfaithful DNA DSB repair will lead to persistent chromosomal damage, promoting altered cell growth and escape from tissue homeostasis. We believe that the proposed approach is highly relevant for assessing the risk of skin cancer in astronauts and for modeling space radiation-induced tumor development in other epithelial tissues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Skin as a model system to study radiation-induced human epithelial cancers: Skin is one of the most complex organs of the body, consisting of multiple cell types, including epithelial keratinocytes and connective tissue (i.e. fibroblasts). Keratinocytes form a stratified squamous epithelium and are composed of one basal cell layer and several differentiated cell layers. Proliferation is limited to the basal layer, whereas differentiating cells withdraw from the cell cycle and express various differentiation markers, dependent on the stage of their differentiation. Because radiation-induced neoplasms in skin derive primarily from epithelial cells, radiobiological studies on keratinocyte monolayers (2D), as suggested by us, provide a good first approximation for assessing cancer development processes triggered by ionizing radiation, but also by other environmental mutagens. Although 2D in vitro assays are high throughput and can be very informative, they do not appropriately recapitulate the complex interactions between different cell types and the extracellular matrix within tissues. The in vivo animal model systems, may provide the complex native environment, but are not well suited for experimental manipulation and may not accurately recapitulate the human condition. Three-dimensional (3D) in vitro organotypic culture models of epithelial tissues, like the one developed by us, provide an intermediate solution by recapitulating the major features of normal 3D tissue organization using human derived cells, and hold great potential for studying the effects of radiation and other environmental mutagens on tissue homeostasis. In this model system, keratinocytes are grown on a collagen layer containing fibroblasts and are exposed to air. These conditions allow fibroblasts and keratinocytes to interact such that a stratified and differentiated epithelium is formed, resembling the architecture of the human skin. For the experiments proposed in our study, we use low-passage normal primary human keratinocytes (NPKs). The advantage of primary keratinocytes is that their response most closely resembles that of normal skin. To mimic intraepithelial neoplastic lesions, we mix NPKs with eGFP fluorescently labeled, immortalized non-tumorigenic HaCaT cells or tumor-derived SCC13 cells. HaCaT cells were derived from normal skin of a male patient and spontaneously immortalized in culture. However, HaCaT cells retain the functional differentiation properties of normal keratinocytes, form a stratified squamous epithelium when grown in organotypic cultures (despite the absence of fully functional p53), and are non-tumorigenic when injected in mice. Notably, HaCaT cells carry a UVB fingerprint mutation in p53 that resides in many skin cells of healthy middle-aged individuals. SCC13 cells were derived from a female, diagnosed with an early stage (T2N0) small facial skin tumor. SCC13 cells also carry a UVB fingerprint mutation in p53 and are characterized by complete dependence on fibroblast feeder support for culture in vitro, and by a non-invasive phenotype in vivo. Thus, while SCC13 cells were derived from a bona fide skin squamous cell carcinoma, the cells cultured from this tumor do not show signs of advanced malignancy. This model for intraepithelial neoplasia is highly relevant since it has been shown previously that adult skin contains clonal patches of p53 mutated keratinocytes approximately 60-3000 cells in size, which can involve up to 4% of the epidermis. We believe that our studies are relevant for a better understanding of the processes involved in the development of epithelial cancers due to exposure to environmental mutagens. The skin model system consisting of co-cultured fibroblasts and keratinocytes is one of the best-understood human tissue systems, and fibroblast-derived soluble factors play an important role in maintaining the well-controlled balance between epithelial proliferation and differentiation. For human skin, many of these paracrine factors and the pathways they are involved in are known, and they also are conserved across other epithelial human organs. Therefore, the response of skin tissue to environmental factors is relevant for predicting cancer risk in other tissues of epithelial origin.
	There is considerable evidence that ionizing radiation causes non-melanoma skin cancer (NMSC), and the risk for NMSC in astronauts exposed to space radiation during long-term space missions is increased [1]. However, the role of high linear energy transfer (LET) ions in oncogenic initiation and progression of skin cells is currently unknown. Here, a human in-vivo-like multi-cellular skin model system is used to further assess the mechanisms that contribute to an increase in skin cancer risk from space radiation. This skin model system, consisting of co-cultured human stromal fibroblasts and keratinocytes [2], is one of the best-understood, reconstructed human tissues, in which stromal-epithelial interactions are critical in maintaining a well controlled balance between epithelial proliferation and differentiation. For human skin, many of the factors and the pathways involved in these interactions are known, and they are also observed across other epithelial human organs. Therefore, we believe that the proposed approach is highly relevant for understanding the mechanisms that contribute to increased risk of skin cancer in astronauts and for modeling space radiation-induced tumor development processes in skin and other epithelial tissues. Multiple different endpoints as surrogate markers for skin cancer are investigated to assess LET and track structure effects. Persistent chromosomal damage (i.e. micronucleus (MN) formation), altered cell proliferation and evasion from apoptosis are assessed in 2D monolayer cultures of normal and initiated human keratinocytes exposed to 300 and 1000 MeV/n Si ions and to 300 and 1000 MeV/n Fe ions, covering an LET range from 44 to 250 keV/μm (Aim 1). All results obtained for the heavy ion exposures are compared to the effects of gamma-rays. In addition, to more realistically recapitulate the complex architecture of a tissue, 3D reconstructed human skin epithelia, composed of fibroblasts and keratinocytes, are used to ascertain the risk of space radiation-induced cancer development processes in 3D (i.e. persistent chromosomal damage, altered cell proliferation, evasion from apoptosis, abnormal cell differentiation and invasion) (Aim 2). To more accurately define skin cancer risk, this 3D model system will be extended further in the future to also include indolent tumor cells (as a surrogate for intraepithelial neoplasia) by mixing tumor cells established from different stages of skin tumor progression with normal keratinocytes in 3D co-cultures. Last, to study the longer-term effects of space radiation exposure on skin cancer processes, a humanized mouse model is employed in

Task Progress:	which irradiated pure or mixed cultures of keratinocytes and tumor cells in 3D epithelia will be grafted onto immunodeficient mice (Aim 3). Late indicators of tumor progression, including tumor invasion, changes to the microenvironment and the presence of human cells in the mouse circulatory system, as an early marker of metastasis, will be assessed. Within the first year of this study we have concentrated on definition experiments, using X-rays at our home institution. We have also participated in one experimental run at NSRL (NSRL11C), in which we irradiated both 2D NPKs and 3D skin epithelia with gamma-rays (for comparison purposes), and used 300 and 1000 MeV/n Si ions, and 300 and 1000 MeV/n Fe ions to assess radiation quality effects. In addition, we have concentrated on further optimizing the growth and the cultivation of 3D skin epithelia, using both NPKs and immortalized non-tumorigenic human HaCaT keratinocytes. We have encountered some difficulties in stetting-up the 3D system with some strains of NPKs, in which we sometimes observed insufficient stratification of the epidermis. For this reason, we decided to conduct our initial studies at BNL with commercially available human EpiDerm full-thickness skin epithelia form MatTek Corporation (Ashland, MA). During the NSRL11C campaign, we have conducted experiments to assess the levels of apoptosis, replication and micronucleus (MN) formation in 2D and 3D NPKs exposed to graded doses of Si and Fe ions, and to gamma-rays. To date, only a small amount of the samples collected has been analyzed. Our preliminary results show that the levels of apoptosis induced in 2D NPKs, for all radiation types and doses (=1 Gy) investigated, are not significantly different from mock-irradiated cells. We conclude that NPKs may induce senescence rather than apoptosis after exposure to ionizing radiation, and this will be investigated further. We find that in 2D NPKs, MN are induced linear with dose (dose range 0.05 – 1 Gy) after exposure to both gamma-rays and 300 MeV/n Si ions, and that the levels of spontaneous and induced MN in NPKs are significantly lower that in non-tumorigenic immortalized keratinocytes. These results suggest a greater tolerance of unrepaired DNA damage in initiated cells compared to normal cells. Last, we assessed the effects of heavy ion exposure on cell proliferation in the basal layer of the epidermis in 3D epithelia by Ki67 immunostaining of paraffin-embedded tissue sections. Compared to mock-irradiated epithelia of the same age, we found a small decrease (~80-85% of control levels) in Ki67-positive cells in tissues exposed to 2 Gy gamma-rays and in tissues exposed to 1 Gy 1000 MeV/n Si ions (LET = 44 keV/µm) at 5 days post exposure. However, compared to mock-irradiated tissues, the amount of Ki67-positive cells in 3D skin epithelia exposed to 1 Gy 1000 MeV/n Fe ions was significantly reduced (35% of control levels), indicating a more drastic effect of this radiation quality with higher LET (151 keV/µm) on cell proliferation. Experiments are underway to also assess a variety of other keratinocyte cell fate decisions in response to these radiation qualities, such as cell differentiation and the recognition of DNA damage and repair. We are currently assessing differentiation-related markers (i.e. loricrin, cytokeratin 10/13) for the stratified exposed and control epithelia, as well as the signature of some proteins involved in the DNA damage response (i.e. 53BP1, gammaH2AX). REFERENCES [1] Kim, M.-H.Y., K. George, and F.A. Cucinotta, Evaluation of skin cancer risk for lunar and Mars missions. Advances in Space Research, 2006. 37: p. 1798-1803. [2] Snijders, A.M., et al., Stromal control of oncogenic traits expressed in response to the overexpression of GLI2, a pleiotropic oncogene. Oncogene, 2009. 28(5): p. 625-37.
Bibliography Type:	Description: (Last Updated: 04/11/2018)
Abstracts for Journals and Proceedings	Snijders AM, Rydberg B, Mannion B, Kronenberg A, Wiese C. "An In-Vivo-Like Multi-Cellular Skin Model System for the Analysis of Radiation Quality Effects." Presented at the 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7087.pdf , Sep-2011